

ROSENDO ADRIÁN DE LOS RIOS MORENO

Robotics & Digital Systems Engineer - ROS2 · Computer Vision · Embedded Systems · Autonomous UAV

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EDUCATION

Tecnológico de Monterrey

Monterrey, México

B.Eng. in Robotics and Digital Systems Engineering - GPA: 91/100

Jun 2026

Relevant Coursework: Advanced Embedded Systems, ROS1/ROS2, Computer Vision, Artificial Intelligence, Autonomous Systems

AIST Tokyo Waterfront - National Institute of Advanced Industrial Science and Technology

Tokyo, Japan

Research Internship - Haptic Intelligence and Robotic Teleoperation

Aug 2025 - Feb 2026

EXPERIENCE

AIST Tokyo Waterfront

Tokyo, Japan

Research Intern - Robotic Teleoperation Systems

Aug 2025 - Feb 2026

- Built an occupancy-aware teleoperation system for a UR5e arm: Intel RealSense D435 → TSDF/ESDF volumetric reconstruction in ROS2 → Unity interface (OpenHRC) → ESDF-based assistive braking at <6 cm from obstacles (40-70 ms braking latency).
- Implemented custom CPU TSDF integrator (Open3D), real-time ESDF node, DBSCAN object clustering, and ROS-TCP-Connector Unity bridge; 15-18 Hz perception pipeline, 140-210 ms end-to-end latency.

Manchester Robotics

Monterrey, México

Training Partner - ROS2 Autonomous Robot

Feb 2025 - Jun 2025

- Built a full ROS2 autonomous control architecture in C++/Python integrating YOLO-trained vision models with real-time navigation; validated through simulation-based unit testing.

John Deere

Monterrey, México

Training Partner - Embedded Systems

Aug 2024 - Dec 2024

- Developed firmware in C/C++ for STM32 microcontrollers with FreeRTOS, integrating IMU sensors via SPI/I2C and motor control via CAN/UART for an autonomous tractor navigation prototype.

PROJECTS

Advanced Oil Pan Tracking & Manipulation System

Feb - May 2025

- ROS2 3D perception pipeline (Kinect v2, Open3D, ICP) for 6-DOF object tracking integrated with xArm robotic arm. Achieved **2.8 cm position accuracy** and **1.8° orientation accuracy** at 450 ms latency.

Vision-Based Autonomous Patrol UAV - Tello Robomaster TT

2024

- Autonomous drone with real-time person detection (OpenCV), multi-threaded vision + flight control architecture, and state machine mission management; integrated djitellopy SDK for low-level UAV commands.

3-DOF Teleoperated Robotic Arm

Jan - Apr 2025

- Real-time human pose detection (MediaPipe, OpenCV) mapped to joint servo commands over ROS2. Achieved **85 ms end-to-end latency** and **<3° angular error** in controlled conditions.

SKILLS

Programming: C++, Python, C, MATLAB, C#

Robotics & Simulation: ROS1, ROS2, Gazebo, NVIDIA Isaac Sim, FreeRTOS - UAV state machines, djitellopy/robomaster SDK

Computer Vision & Perception: OpenCV, YOLO, MediaPipe, Open3D, ICP Algorithm, point cloud processing

Hardware & Protocols: STM32, Raspberry Pi, xArm, Arduino - CAN, SPI, UART, I2C, TCP/IP

DevOps & Tools: Git, Docker, VS Code, Linux

Languages: Spanish (Native) · English B2 - TOEFL IBT

CERTIFICATIONS

TOEFL IBT - English B2 Certificate

2024